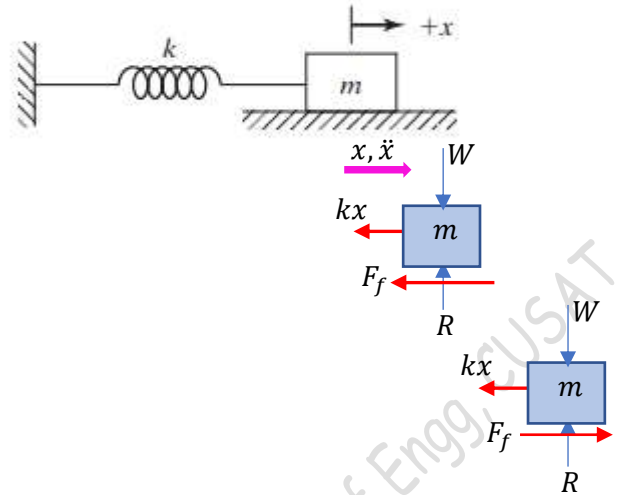


Free vibration with Coulomb damping

Consider a single degree of freedom spring mass system with Coulomb damping. The damping force here will be the dry friction existing between the block and the surface. Depending upon the direction of motion, the direction of frictional force will be different. When the mass moves from the **left extreme to the right extreme**, the direction of frictional force ($F_f = \mu R = \mu W$) is towards **left**, and when the mass moves from **right extreme to the left extreme**, the direction of friction (F_f) is towards **right**.



Let us consider the free-body diagram of the mass when it moves to the right of the equilibrium position to the right extreme ($L \rightarrow R$).

$$m\ddot{x} = -kx - F_f$$

When it returns and the mass is still on the right side of the equilibrium position ($R \rightarrow L$),

$$m\ddot{x} = -kx + F_f$$

The two equations can be combined as,

$$m\ddot{x} + kx = \mp F_f$$

This is a second order, non-homogeneous, linear differential equation. The solution will have two parts; a homogeneous solution and a particular solution.

Homogeneous solution

$$m\ddot{x} + kx = 0 \quad \langle\langle\rangle\rangle \quad x_h = X_0 \sin(\omega_n t + \varphi)$$

Particular solution: Since the right hand side of the differential equation is a constant, let us consider the particular solution also as a constant.

$$x_p = C$$

Substituting this value of particular solution in to the differential equation

$$m \times 0 + kC = \mp F_f \quad \langle\langle\rangle\rangle \quad C = \frac{\mp F_f}{k}$$

$$x_p = \frac{\mp F_f}{k}$$

The **complete solution** is

$$x(t) = X_0 \sin(\omega_n t + \varphi) \mp \frac{F_f}{k}$$

The velocity is given by, $\dot{x}(t) = X_0 \omega_n \cos(\omega_n t + \varphi)$

Assume the initial conditions as $x(t=0) = x_0$ and $\dot{x}(t=0) = 0$. That is, the mass is pulled to the right x_0 and released. The mass moves from ($R \rightarrow L$)

$$x(t) = X_0 \sin(\omega_n t + \varphi) + \frac{F_f}{k} \quad \langle\langle\rangle\rangle \quad \dot{x}(t) = X_0 \omega_n \cos(\omega_n t + \varphi)$$

$$x_0 = X_0 \sin(\varphi) + \frac{F_f}{k} \quad \langle\langle \text{and} \rangle\rangle \quad 0 = X_0 \omega_n \cos(\varphi) \rightarrow \varphi = 90^\circ$$

$$X_0 = x_0 - \frac{F_f}{k} \quad \langle\langle\langle\rangle\rangle\rangle \quad x(t) = \left(x_0 - \frac{F_f}{k}\right) \cos \omega_n t + \frac{F_f}{k} \quad (R \rightarrow L)$$

When $t = \frac{\pi}{\omega_n}$ (half the time period), the mass reaches the left extreme. The position of the mass at the left extreme is given by,

$$x(t = \pi/\omega_n) = \left(x_0 - \frac{F_f}{k}\right) \cos(\omega_n \times \frac{\pi}{\omega_n}) + \frac{F_f}{k} = -\left(x_0 - \frac{F_f}{k}\right) + \frac{F_f}{k} = -\left(x_0 - \frac{2F_f}{k}\right)$$

Now, the mass moves from (L → R) with initial displacement = $-\left(x_0 - \frac{2F_f}{k}\right)$ and initial velocity = 0.

$$x(t) = X_0 \sin(\omega_n t + \varphi) - \frac{F_f}{k} \quad \ll \gg \quad \dot{x}(t) = X_0 \omega_n \cos(\omega_n t + \varphi)$$

$$-\left(x_0 - \frac{2F_f}{k}\right) = X_0 \sin(\varphi) - \frac{F_f}{k} \quad \ll \text{ and } \gg \quad 0 = X_0 \omega_n \cos(\varphi) \rightarrow \varphi = 90^\circ$$

$$X_0 = -\left(x_0 - \frac{3F_f}{k}\right) \quad \ll \ll \gg \gg \quad x(t) = -\left(x_0 - \frac{3F_f}{k}\right) \cos \omega_n t - \frac{F_f}{k} \quad (L \rightarrow R)$$

When $t = \frac{\pi}{\omega_n}$ (half the time period), the mass reaches the right extreme. The position of the mass at the right extreme is given by,

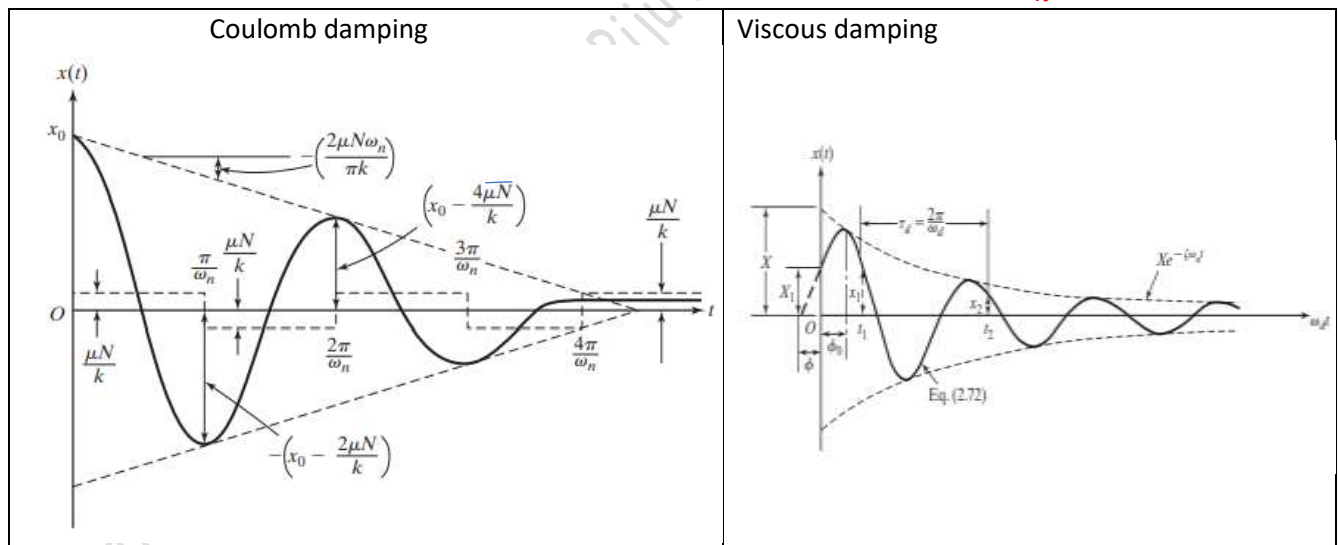
$$x(t = \pi/\omega_n) = -\left(x_0 - \frac{3F_f}{k}\right) \cos(\omega_n \times \frac{\pi}{\omega_n}) - \frac{F_f}{k} = \left(x_0 - \frac{3F_f}{k}\right) - \frac{F_f}{k} = +\left(x_0 - \frac{4F_f}{k}\right)$$

Hence, the amplitude of vibration decreases by $\frac{4F_f}{k}$ after every cycle, or by $\frac{2F_f}{k}$ in every half cycle.

This will continue till the n^{th} half cycle when the displacement $x_0 - n \frac{2F_f}{k}$ produces a spring force $k\left(x_0 - n \frac{2F_f}{k}\right)$ which is less than the frictional force F_f .

The number of half cycles (n) before the mass comes to stationary can be found out by,

$$k\left(x_0 - n \frac{2F_f}{k}\right) \leq F_f \quad \ll \gg \quad \left(x_0 - n \frac{2F_f}{k}\right) \leq \frac{F_f}{k} \quad \ll \gg \quad n \geq \frac{\left(x_0 - \frac{F_f}{k}\right)}{\frac{2F_f}{k}}$$



Compare the responses !!!!!

Example 1

A metal block, placed on a rough surface, is attached to a spring and is given an initial displacement of 10 cm from its equilibrium position. After 5 cycles of oscillation in 2 s, the final position of the metal block is found to be 1 cm from its equilibrium position. Find the coefficient of friction between the surface and the metal block.

$$x_0 = 0.1 \text{ m} \quad \lll \ggg \quad \text{period of oscillation, } T = \frac{2}{5} = 0.4 \text{ s} \quad \lll \ggg \quad \omega_n = \frac{2\pi}{T} = 15.7 \text{ rad/s}$$

$$\omega_n = \sqrt{\frac{k}{m}} \ggg k = 246.5 \text{ m}$$

No. of half cycles before it comes to rest, $n = 10$.

Final stopping position after 10 half cycles = 0.01 m

$$\begin{aligned} \text{i.e., } x_0 - n \frac{2F_f}{k} &= 0.01 \\ 0.1 - \frac{10 \times 2 \times F_f}{k} &= 0.01 \ggg 0.1 - \frac{10 \times 2 \times \mu mg}{246.5m} = 0.01 \ggg \mu = 0.113 \end{aligned}$$

#Example 2

A steel shaft of length 1m and diameter 50 mm is fixed at one end and carries a pulley of mass moment of inertia 25 kg.m² at the other end. A band brake exerts a constant frictional torque of 400 Nm around the circumference of the pulley. If the pulley is displaced by 6° and released, (1) determine the number of cycles before the pulley comes to rest and (2) the final settling position of the pulley. Take $G = 80 \text{ GPa}$

$$\text{The torsional stiffness of the shaft, } k = \frac{GJ}{l} = \frac{80 \times 10^9 \times \frac{\pi}{32} \times 0.05^4}{1} = 49087.4 \text{ Nm/rad}$$

$$\text{Mass moment of inertia of the disc, } I = 25 \text{ kg.m}^2$$

$$\text{Frictional torque, } T_f = 400 \text{ Nm}$$

$$\text{Initial angular displacement, } \theta_0 = 6^\circ = 0.105 \text{ rad}$$

Number of half cycles before the system comes to rest,

$$n \geq \frac{\left(x_0 - \frac{F_f}{k}\right)}{\frac{2F_f}{k}} \geq \frac{\left(\theta_0 - \frac{T_f}{k}\right)}{\frac{2T_f}{k}} \geq \frac{\left(0.105 - \frac{400}{49087.4}\right)}{\frac{2 \times 400}{49087.4}} \geq 5.94 \cong 6$$

The number of cycles $\cong 3$

The final angular position after 6 half cycles,

$$\theta = \left(\theta_0 - 6 \times \frac{2 \times T_f}{k}\right) = 0.105 - 6 \times \frac{2 \times 400}{49087.4} = 0.0072 \text{ rad} = 0.4^\circ$$

Exercise 1

A metal block, placed on a rough surface, is attached to a spring and is given an initial displacement of 10 cm from its equilibrium position. It is found that the natural time period of motion is 1 s and that the amplitude reduces by 0.5 cm in each cycle. Find the kinetic coefficient of friction between the metal block and the surface and the number of cycles of motion executed by the block before it stops.

#Exercise 2

A single degree of freedom system consists of a mass 20 kg and a spring of stiffness 4000 N/m . The amplitudes of successive cycles are found to be $50, 45, 40, 35, \dots$ mm. Determine the nature and magnitude of the damping force and the frequency of damped vibrations.

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition